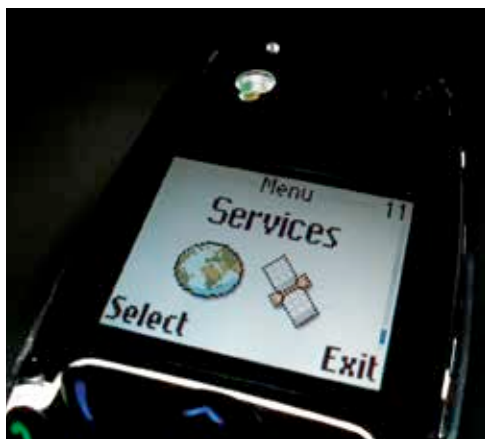


At this level, the technology is good for small scale electronics like calculators and digital wrist watches. But to be potentially useful for larger displays like that of televisions and computers, the technology was improved to be able to handle colour filtration instead of just dealing with white light. Now, along with filtering the amount of light passing through, each pixel could also use its orientation to define the colour being displayed on the screen.

TN derivatives and evolutions

One of the major milestones in TN technology was **Super Twisted Nematic Displays**, or **STNs**. Standard TN displays didn't have much of a difference between the ON and the OFF thresholds for the pixels, due to the low angle of 90 degrees between the two states. In the mid 1980's, STN displays were invented which solved this problem by allowing the light director to rotate through an angle of 270 degrees.

On a smaller display, like those in calculators and watches, to control each segment voltage can be directly applied to it through a direct connection. But for any larger displays, this would mean millions of connections to cover all the pixels. To resolve this, pixels are addressable with rows and columns, bringing down the connections from millions to thousands. To control each pixel on the display, the display needs to use a multiplexed scheme of addressing that can deal with a large number of pixels at the same time. The lower difference between the ON and OFF voltage of the STN made this possible.



A 12 Bit CSTN on a Nokia 3510i

To further reduce undesirable coloration due to a shifted transmission spectrum, **Double STN**, or **D-STN** was implemented. As the name sounds, it consisted of two STN layers in a single cell with opposite twists to each other to negate undesirable phase shifts in the light.

Another type of STN display was the